

TECHNICAL SKILLS

Proficient Languages: Python, R, SQL

Tools & Libraries: Jupyter Notebook (Anaconda), Sklearn, ggplot, Excel, Git, LaTeX

Machine Learning & Stats: KNN, K-Means Clustering, PCA, GLMs, EM-Algorithm

EDUCATION

Master of Mathematics and Statistics (MMathStat): 1st Class (73% so far), *University of Warwick* July 2027

Relevant Modules: Maths of Machine Learning, Machine Learning Frameworks, Multivariate Statistics, Generalised Linear Models, Python for Data-Analytic Tasks, Applied Statistical Modelling, Probability Theory

Mathematics A*, Further Mathematics A*, Economics A, *Notre Dame High School - Sheffield* Sep 2023

PROJECTS

Premier League Stratification Analysis | *R / Excel / LaTeX*

- Engineered a machine learning pipeline to challenge the "Big 6" narrative by extracting and analysing multi-dimensional team performance metrics.
- Applied Principal Component Analysis (PCA) for dimensionality reduction and K-Means clustering to identify hidden similarities and group teams into distinct statistical profiles.

Car Price Prediction Engine | *R / LaTeX*

- Conducted Exploratory Data Analysis (EDA) on a real-world Kaggle dataset, utilising K-Nearest Neighbours (KNN) to robustly impute missing data points.
- Developed and tuned Ridge, Lasso, and Stepwise regression models to predict car prices, evaluating accuracy and preventing overfitting using R^2 and Adjusted R^2 metrics.

Fantasy Football Analytics Pipeline | *Excel*

- Automated a data pipeline using advanced Excel functions (e.g., XLOOKUP, dynamic arrays) to aggregate statistics, generate weekly metrics, and manage a custom league for the University Statistics team.

EXPERIENCE

Barnett Waddingham **Leeds, UK**
Actuarial Consulting Intern July – Aug 2025

- Supported actuarial valuations for pension schemes with 5,000+ members by validating 10+ cash equivalent transfer values and reconciling member data using Excel.
- Produced and updated multiple client actuarial reports and accounting disclosures, ensuring accuracy and delivering work to tight deadlines.
- Completed structured assignments including drafting professional member communications and presenting funding valuations to mock trustees, strengthening analytical and communication skills.

Sheffield Scitt **Sheffield, UK**
Maths Teacher Intern Jun – Jul 2024

- Applied diverse teaching techniques to maintain engagement across varied ability groups.
- Led tutor groups of 5–10 students, supporting a 50% improvement in student test scores.

INTERESTS

- Active member of the University Statistics Football team, having played competitively since childhood. Keen squash player and eager to try new sports.
- Grade 5 pianist with a particular passion for learning cinematic film scores, and an enthusiastic home cook who enjoys experimenting with new recipes.